Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Parallel lines and transversals**

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 14 cm. Multiplying the original length by the scale factor gives $7 \times 2 = 14$ cm.

2 **[Foundational]** • **Target: Parallel lines and transversals**

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 9. Substituting $x = 1$ into $y = 3x + 6$ gives $y = 3(1) + 6 = 9$.

3 **[Foundational]** • **Target: Parallel lines and transversals**

In ΔABC studying parallel lines and transversals, $m\angle A = 53^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 75°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (53^\circ + 52^\circ) = 75^\circ$.

4 **[Foundational]** • **Target: Parallel lines and transversals**

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 21 cm. Multiplying the original length by the scale factor gives $7 \times 3 = 21$ cm.

5 **[Foundational]** • **Target: Parallel lines and transversals**

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 12. Substituting $x = 2$ into $y = 3x + 6$ gives $y = 3(2) + 6 = 12$.

6 **[Foundational]** • **Target: Parallel lines and transversals**

In ΔABC studying parallel lines and transversals, $m\angle A = 53^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 68°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (53^\circ + 59^\circ) = 68^\circ$.

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7 [Foundational] • Target: Parallel lines and transversals

A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 28 cm. Multiplying the original length by the scale factor gives $7 \times 4 = 28$ cm.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Parallel lines and transversals

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 15. Substituting $x = 3$ into $y = 3x + 6$ gives $y = 3(3) + 6 = 15$.

9 [Intermediate] • Target: Parallel lines and transversals

In ΔABC studying parallel lines and transversals, $m\angle A = 53^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 61°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (53^\circ + 66^\circ) = 61^\circ$.

10 [Intermediate] • Target: Parallel lines and transversals

A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 35 cm. Multiplying the original length by the scale factor gives $7 \times 5 = 35$ cm.

11 [Intermediate] • Target: Parallel lines and transversals

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Final Answer: 18. Substituting $x = 4$ into $y = 3x + 6$ gives $y = 3(4) + 6 = 18$.

12 [Intermediate] • Target: Triangles

In ΔABC studying triangles, $m\angle A = 54^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 81°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (54^\circ + 45^\circ) = 81^\circ$.

13 [Intermediate] • Target: Triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 16 cm. Multiplying the original length by the scale factor gives $8 \times 2 = 16$ cm.

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14 [Intermediate] • Target: Triangles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 11. Substituting $x = 1$ into $y = 4x + 7$ gives $y = 4(1) + 7 = 11$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Triangles • Cross-Lesson: 5.01 Geometry

In ΔABC studying triangles, $m\angle A = 54^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 74°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (54^\circ + 52^\circ) = 74^\circ$.

16 [Challenging] • Target: Triangles • Cross-Lesson: 6.01 Trigonometry

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 24 cm. Multiplying the original length by the scale factor gives $8 \times 3 = 24$ cm.

17 [Challenging] • Target: Triangles • Cross-Lesson: 7.01 Transformations & Vectors

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 15. Substituting $x = 2$ into $y = 4x + 7$ gives $y = 4(2) + 7 = 15$.

18 [Challenging] • Target: Triangles • Cross-Lesson: 8.01 Probability & Statistics

In ΔABC studying triangles, $m\angle A = 54^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 67°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (54^\circ + 59^\circ) = 67^\circ$.

19 [Challenging] • Target: Triangles • Cross-Lesson: 1.01 Number

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 32 cm. Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

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20 [Challenging] • Target: Triangles • Cross-Lesson: 10.01 Sets & Venn Diagrams

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 19. Substituting $x = 3$ into $y = 4x + 7$ gives $y = 4(3) + 7 = 19$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Geometrical diagrams

In ΔABC studying geometrical diagrams, $m\angle A = 51^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 84° ✓

b. 94°

c. 76°

d. 129°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 45^\circ) = 84^\circ$.

22 [Foundational] • Target: Geometrical diagrams

A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?

a. 12 cm

b. 15 cm ✓

c. 20 cm

d. 45 cm

Multiplying the original length by the scale factor gives $5 \times 3 = 15$ cm.

23 [Foundational] • Target: Geometrical diagrams

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?

a. 14

b. 21

c. 19 ✓

d. 24

Substituting $x = 3$ into $y = 5x + 4$ gives $y = 5(3) + 4 = 19$.

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24 [Foundational] • Target: Geometrical diagrams

In ΔABC studying geometrical diagrams, $m\angle A = 51^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 63° ✓

b. 73°

c. 55°

d. 129°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 66^\circ) = 63^\circ$.

25 [Foundational] • Target: Measuring angles

In ΔABC studying measuring angles, $m\angle A = 52^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 83° ✓

b. 93°

c. 75°

d. 128°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (52^\circ + 45^\circ) = 83^\circ$.

26 [Foundational] • Target: Measuring angles

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm ✓

c. 24 cm

d. 54 cm

Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

27 [Foundational] • Target: Measuring angles

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$?

a. 9

b. 13

c. 11 ✓

d. 13

Substituting $x = 3$ into $y = 2x + 5$ gives $y = 2(3) + 5 = 11$.

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28 [Foundational] • Target: Measuring angles

In ΔABC studying measuring angles, $m\angle A = 52^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 62° ✓

b. 72°

c. 54°

d. 128°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (52^\circ + 66^\circ) = 62^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Geometrical diagrams

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm ✓

c. 15 cm

d. 20 cm

Multiplying the original length by the scale factor gives $5 \times 2 = 10$ cm.

30 [Intermediate] • Target: Geometrical diagrams

In ΔABC studying geometrical diagrams, $m\angle A = 51^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 77° ✓

b. 87°

c. 69°

d. 129°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 52^\circ) = 77^\circ$.

31 [Intermediate] • Target: Geometrical diagrams

A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm ✓

c. 25 cm

d. 80 cm

Multiplying the original length by the scale factor gives $5 \times 4 = 20$ cm.

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32 [Intermediate] • Target: Geometrical diagrams

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 19

b. 26

c. 24 ✓

d. 29

Substituting $x = 4$ into $y = 5x + 4$ gives $y = 5(4) + 4 = 24$.

33 [Intermediate] • Target: Measuring angles

In ΔABC studying measuring angles, $m\angle A = 52^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 76° ✓

b. 86°

c. 68°

d. 128°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (52^\circ + 52^\circ) = 76^\circ$.

34 [Intermediate] • Target: Measuring angles

In ΔABC studying measuring angles, $m\angle A = 52^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 69° ✓

b. 79°

c. 61°

d. 128°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (52^\circ + 59^\circ) = 69^\circ$.

35 [Intermediate] • Target: Measuring angles

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$?

a. 11

b. 15

c. 13 ✓

d. 15

Substituting $x = 4$ into $y = 2x + 5$ gives $y = 2(4) + 5 = 13$.

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36 [Intermediate] • Target: Parallel lines and transversals

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 1$?

a. 6

b. 11

c. 9 ✓

d. 12

Substituting $x = 1$ into $y = 3x + 6$ gives $y = 3(1) + 6 = 9$.

37 [Intermediate] • Target: Parallel lines and transversals

In ΔABC studying parallel lines and transversals, $m\angle A = 53^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 68° ✓

b. 78°

c. 60°

d. 127°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (53^\circ + 59^\circ) = 68^\circ$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Parallel lines and transversals • Cross-Lesson: 8.01 Probability & Statistics

A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm ✓

c. 42 cm

d. 175 cm

Multiplying the original length by the scale factor gives $7 \times 5 = 35$ cm.

39 [Challenging] • Target: Triangles • Cross-Lesson: 1.01 Number

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 13

c. 11 ✓

d. 15

Substituting $x = 1$ into $y = 4x + 7$ gives $y = 4(1) + 7 = 11$.

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40 [Challenging] • Target: Triangles • Cross-Lesson: 10.01 Sets & Venn Diagrams

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$?

a. 11

b. 17

c. 15 ✓

d. 19

Substituting $x = 2$ into $y = 4x + 7$ gives $y = 4(2) + 7 = 15$.

41 [Challenging] • Target: Triangles • Cross-Lesson: 2.01 Algebra & Graphs

A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm

b. 40 cm ✓

c. 48 cm

d. 200 cm

Multiplying the original length by the scale factor gives $8 \times 5 = 40$ cm.

42 [Challenging] • Target: Quadrilaterals • Cross-Lesson: 3.01 Coordinate Geometry

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ **and** $x = 6$ ✓

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

43 [Challenging] • Target: Quadrilaterals • Cross-Lesson: 4.01 Mensuration

For the model $f(x) = 2x + 8$ in quadrilaterals, evaluate $f(4)$.

a. 16 ✓

b. 18

c. 8

d. 20

Substitute $x = 4$: $f(4) = 2(4) + 8 = 16$.

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44 [Challenging] • Target: Quadrilaterals • Cross-Lesson: 5.01 Geometry

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ **and** $x = 9$ ✓

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

45 [Challenging] • Target: Angles of polygons • Cross-Lesson: 6.01 Trigonometry

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$?

a. 11

b. 15

c. 13 ✓

d. 15

Substituting $x = 2$ into $y = 2x + 9$ gives $y = 2(2) + 9 = 13$.

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